



Tag 2

23.06.2026

AZ-900

Learning path 1:
Cloud concepts

Guten Morgen!



- Microsoft Learn Account

Login

- Skillable Lab Umgebung

Login

AZ-900

AZ-104

bis Dezember

- Measure Up Prüfungssimulation

Login

Prinzipien

- Theoria cum Praxi

- Github A5

- Englisch

Github Repo A5

<https://github.com/www42/a5>

Links

Agenda

Präsentationen

Agenda

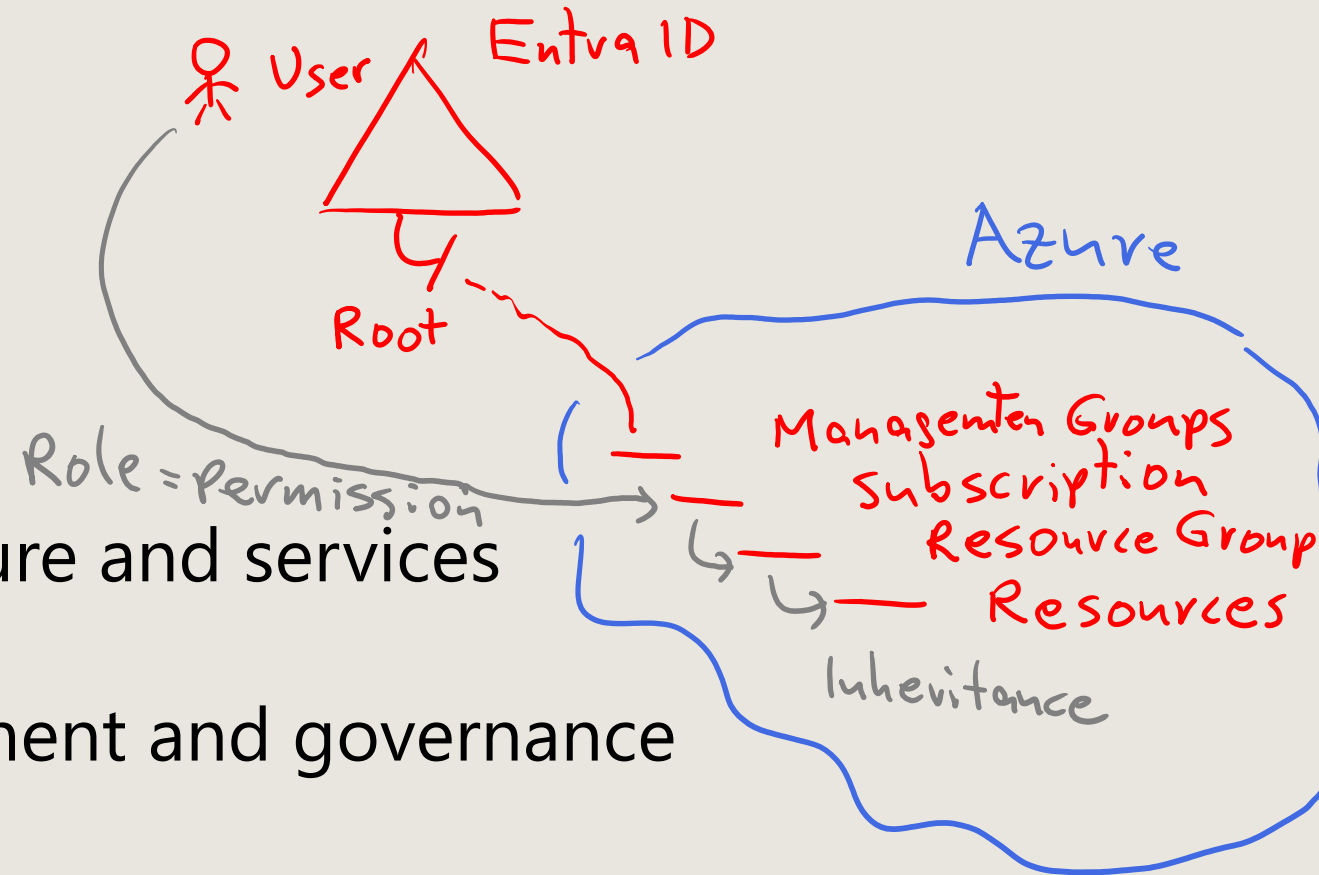
AZ-900

Labs
(Guided Projects)

Learning path 1: Cloud concepts

Learning path 2: Azure architecture and services

Learning path 3: Azure management and governance



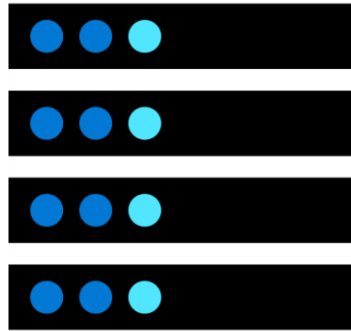
Module 1: Describe cloud computing



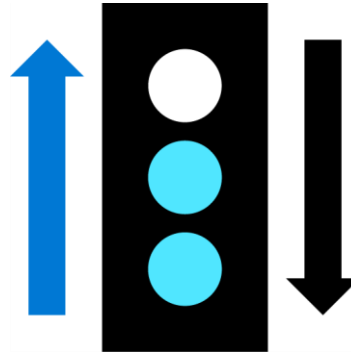
What is cloud computing?

Kaufhaus

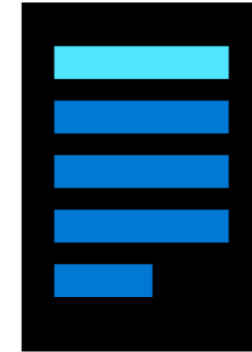
Cloud computing is the delivery of computing services over the internet, enabling faster innovation, flexible resources, and economies of scale.



Compute



Networking

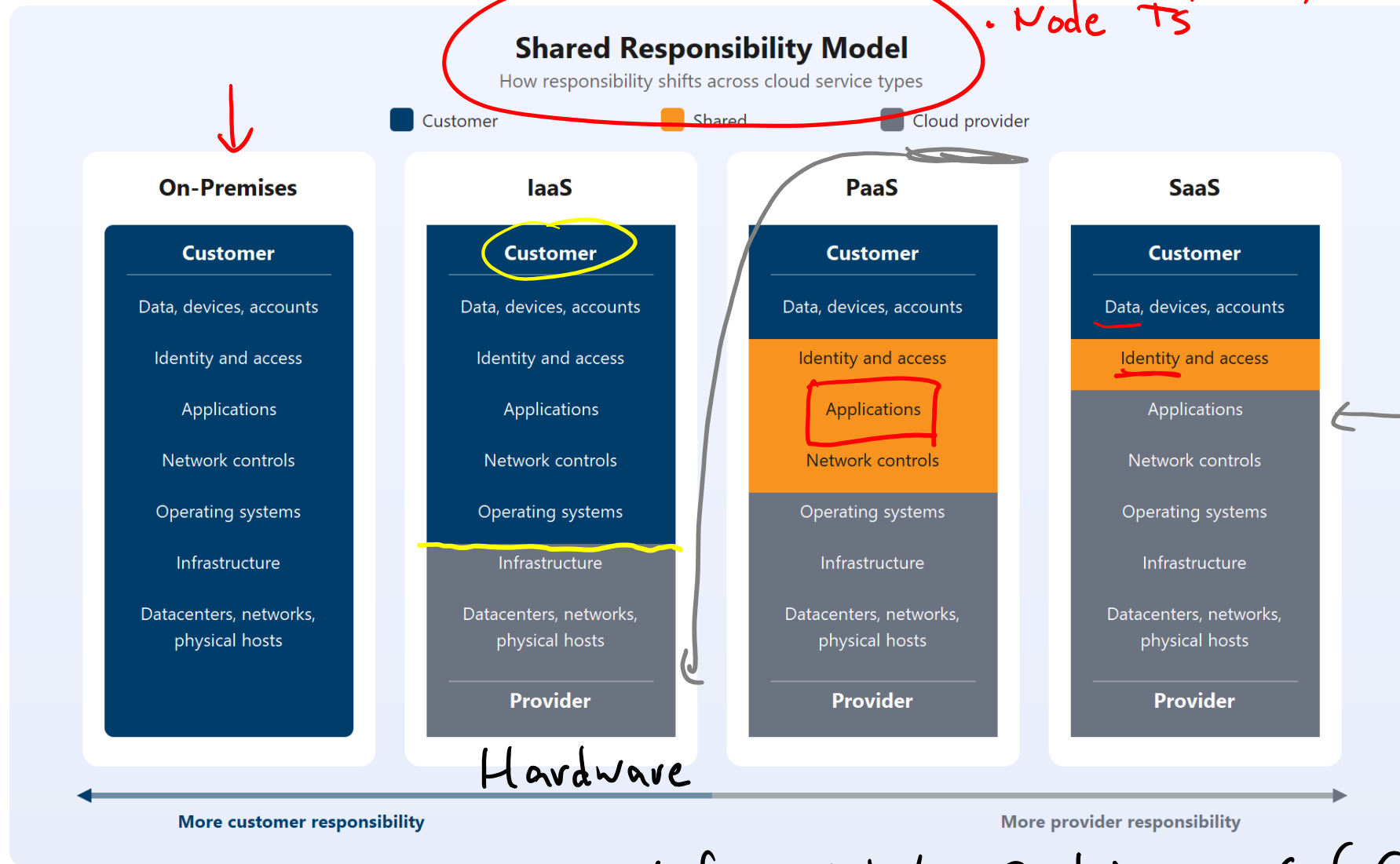


Storage

The shared responsibility model

Platform
z.B. Python
.Net (C#, C++, Q#...)
.Node TS

App Service



NIST
Paper
2011

Photoshop

Adobe

Zwischenstufen möglich: ContaineraaS (CaaS)

Cloud deployment models

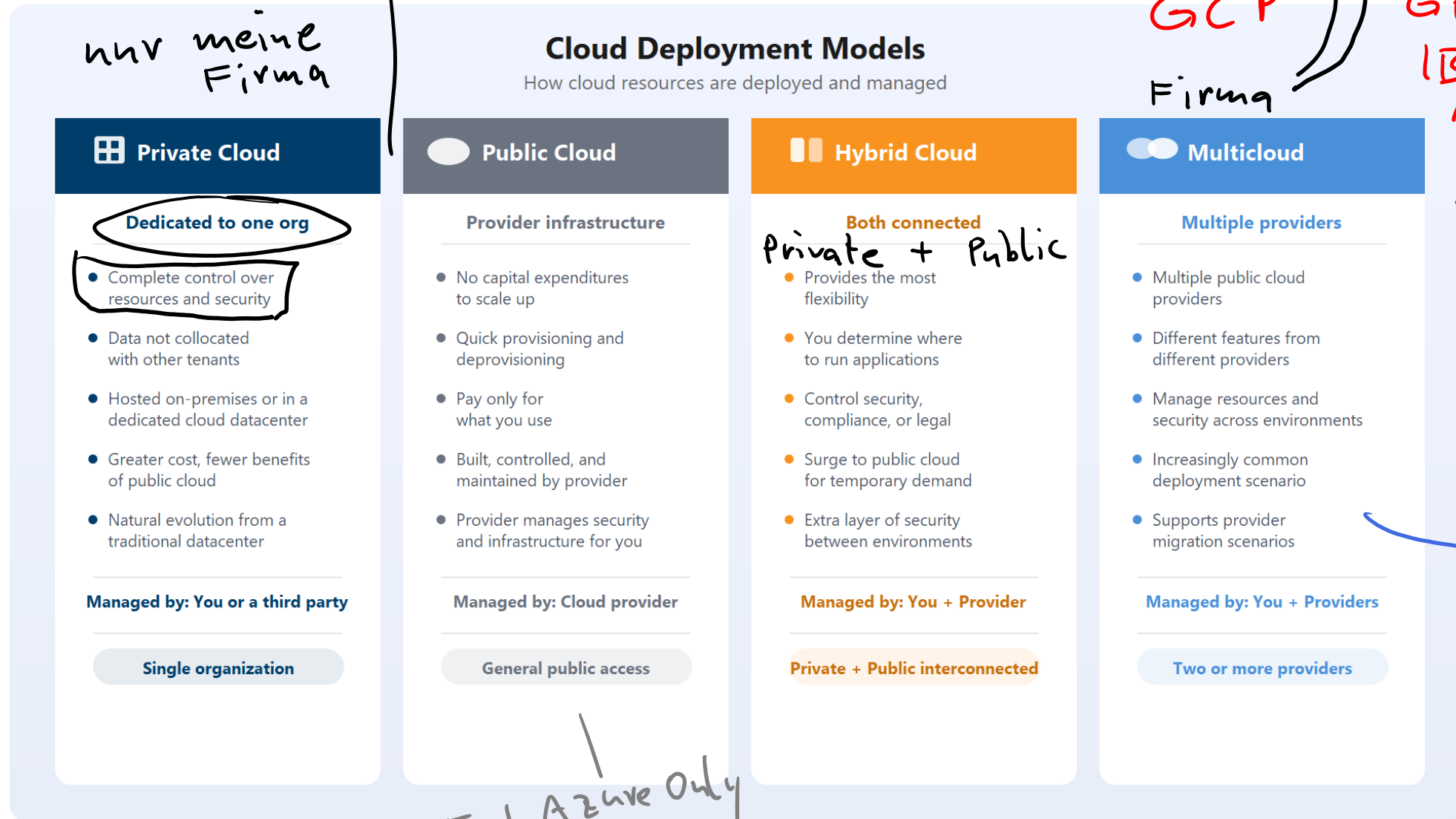
nur 1 Provider

AWS
Azure
GCP
Firma

Amazon
Microsoft
Google
IBM
Alibaba
Digital Ocean
↳ Open Cloud

On Prem
auf
höherem
Level

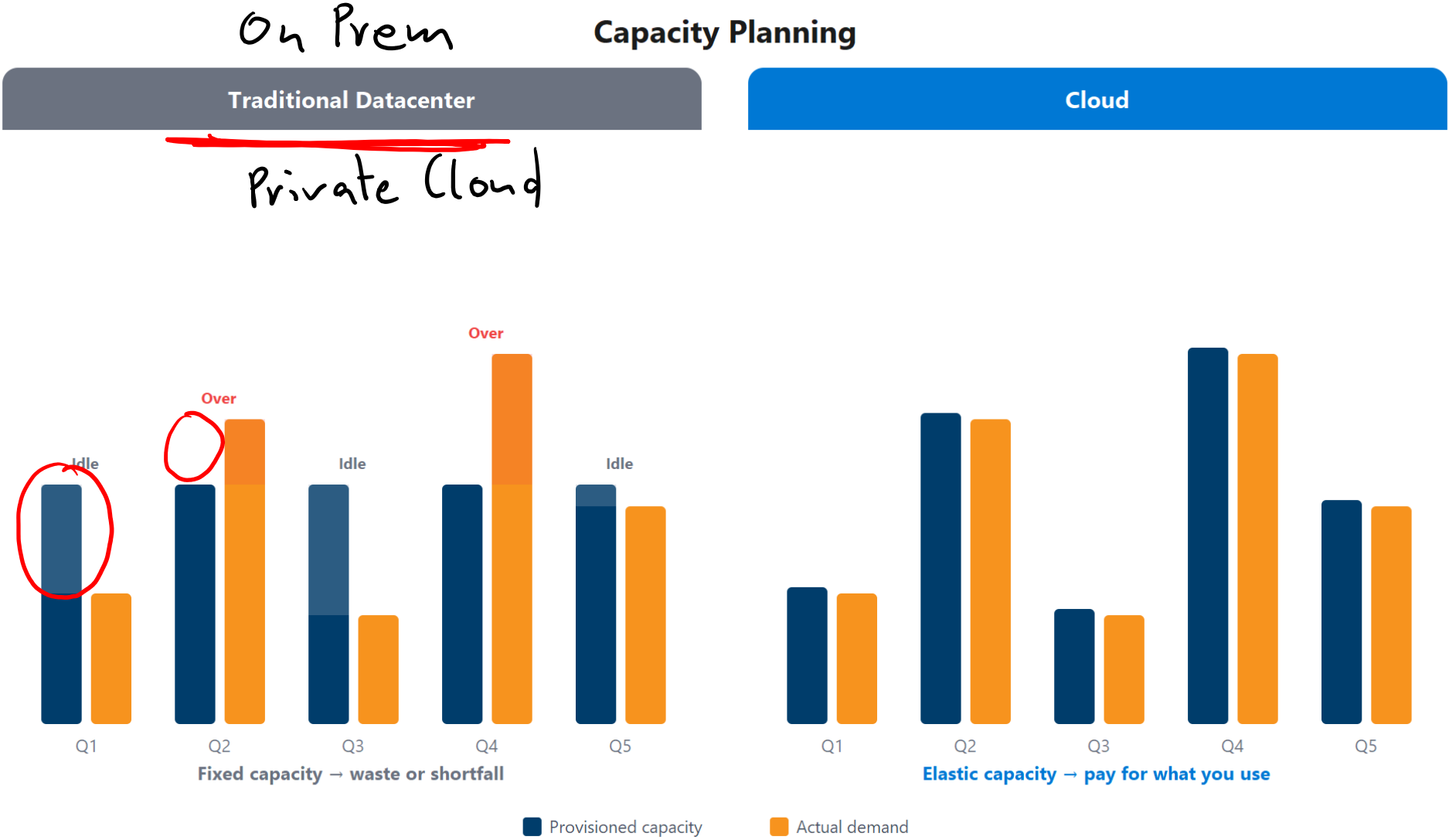
nur meine
Firma



Tool Azure only
Bicep

Tool
Terraform
(HCL)

The consumption-based model



Knowledge check



1 In the shared responsibility model, which responsibility always remains with the customer across IaaS, PaaS, and SaaS?

- ☐ Physical hosts
- ☐ Datacenter cooling and power
- ☒ Data and identities

Module 2: Describe the benefits of using cloud services



High availability and scalability

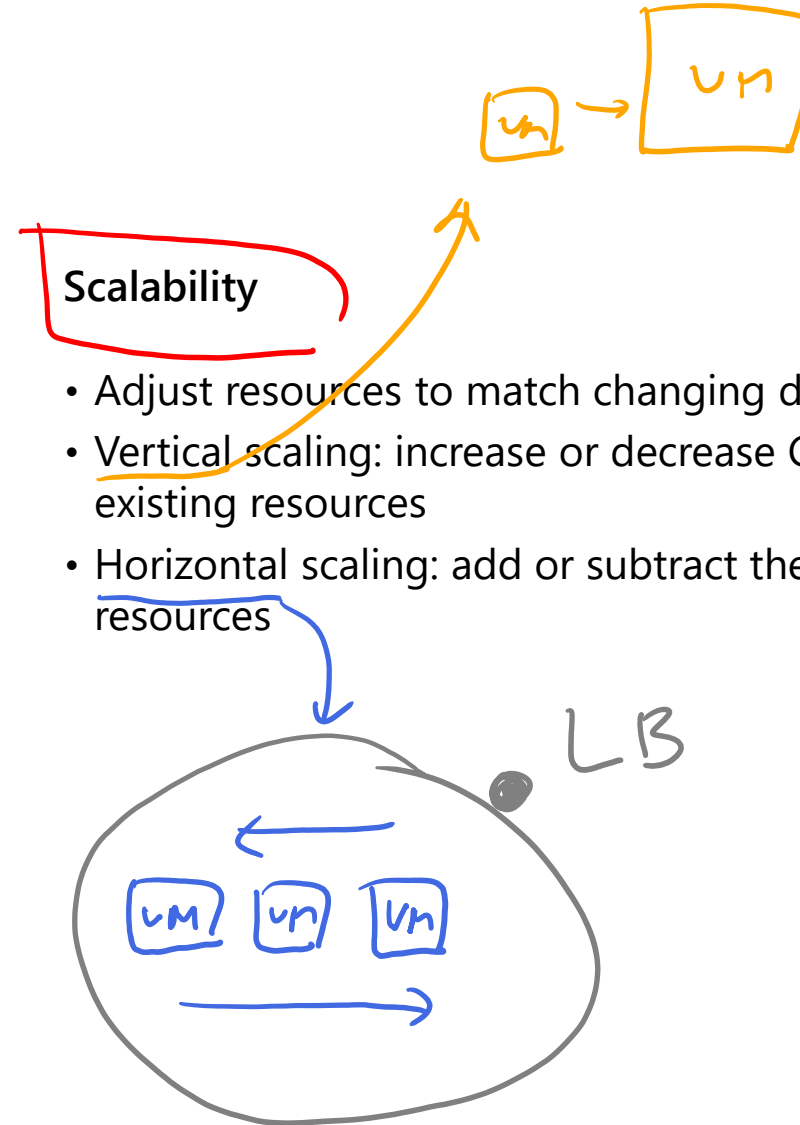
High availability

HA LRS
ZRS
GRS

- Focus on maximum availability regardless of disruptions or events
- Azure SLAs define uptime guarantees for each service
- Architect solutions with high availability in mind

Scalability

- Adjust resources to match changing demand
- Vertical scaling: increase or decrease CPU/RAM of existing resources
- Horizontal scaling: add or subtract the number of resources



Reliability and predictability

Reliability

SLA

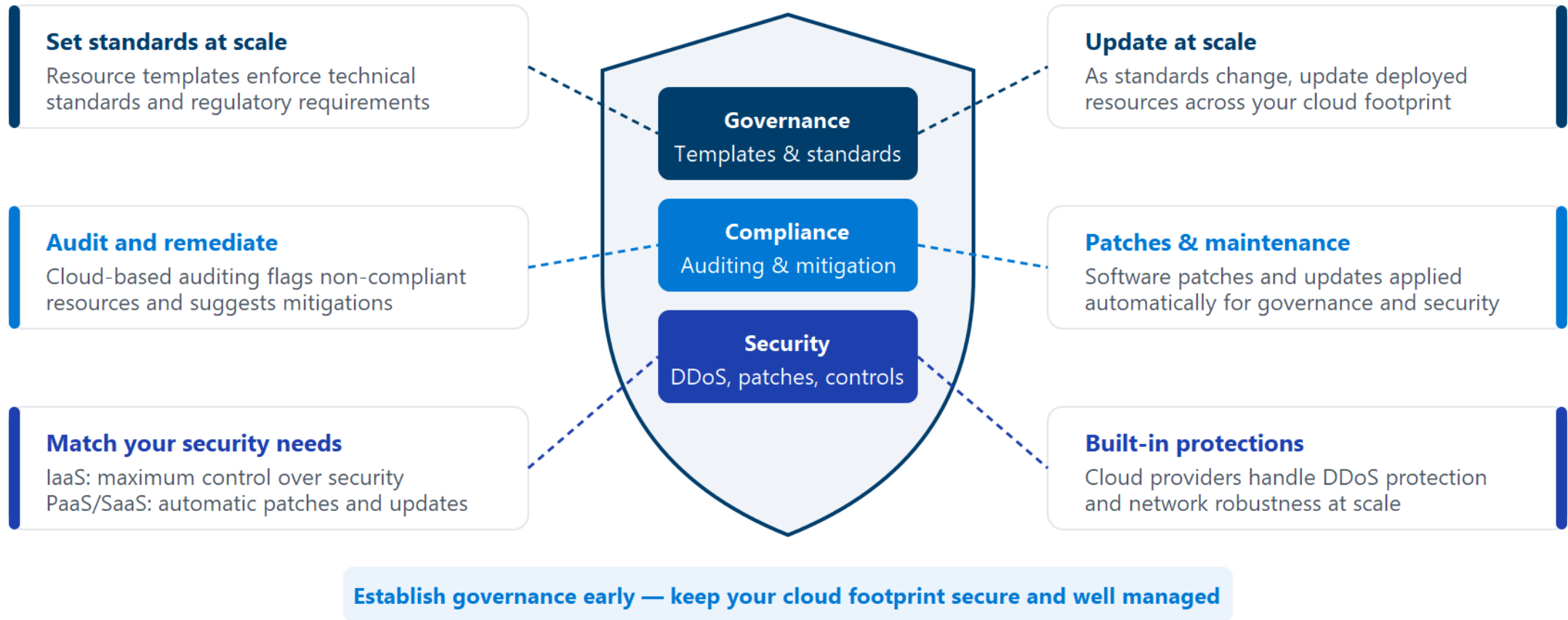
- Ability of a system to recover from failures and continue to function
- Part of the Microsoft Azure Well-Architected Framework
- Decentralized design enables deployment in regions around the world
- Automatic failover to a different region in the event of catastrophe

Predictability

Cost

- Performance predictability: predict resources needed for a good experience
- Cost predictability: predict or forecast the cost of cloud spend
- Azure Pricing Calculator helps estimate cloud cost

Security and governance in the cloud



Manageability in the cloud

Management of the cloud

- Automatically scale resource deployment based on need
- Deploy resources based on preconfigured templates
- Monitor the health of resources and automatically replace failures
- Receive automatic alerts based on configured metrics

Management in the cloud

- Through a web portal
- Using a command-line interface (CLI)
- Using APIs
- Using PowerShell

Knowledge check



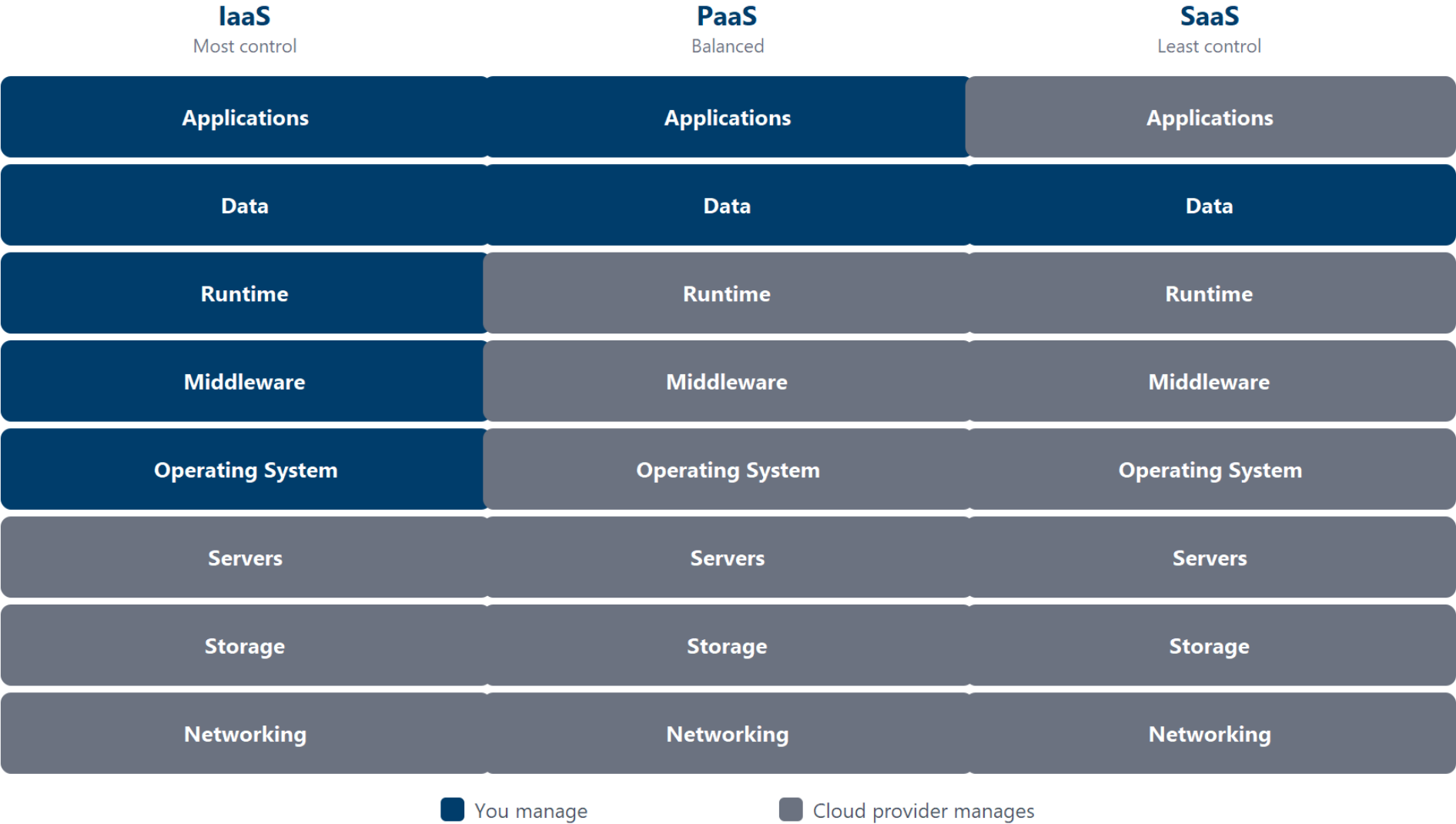
1 Which cloud benefit is best demonstrated by automatically adding resources during peak demand and removing them later?

- ☐ Reliability
- ☒ Elasticity
- ☐ Governance

Module 3: Describe cloud service types

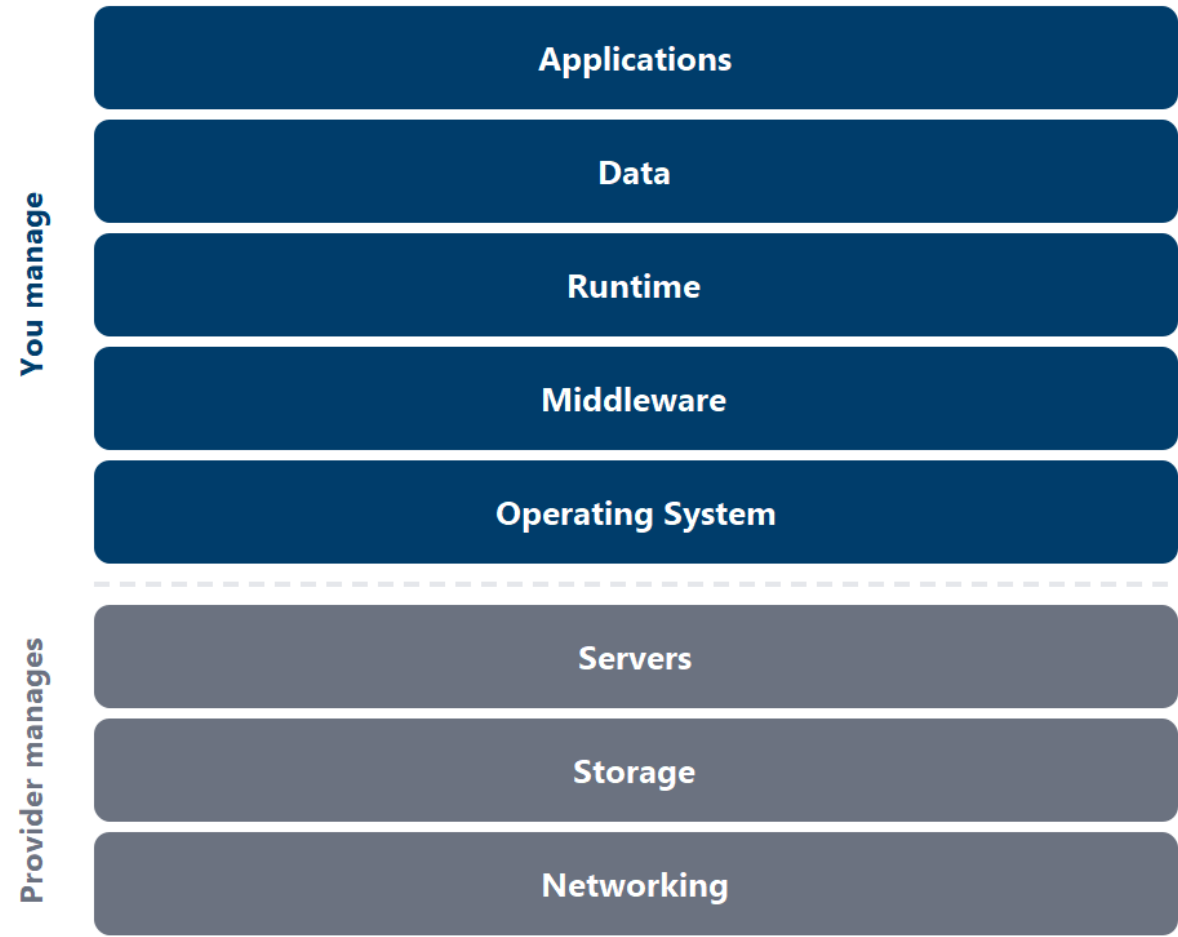


Cloud service types comparison



Infrastructure as a service (IaaS)

Maximum control — you manage the workload stack



Common Scenarios

Lift-and-shift migration

Set up cloud resources similar to on-premises, then move workloads to IaaS infrastructure.

Testing and development

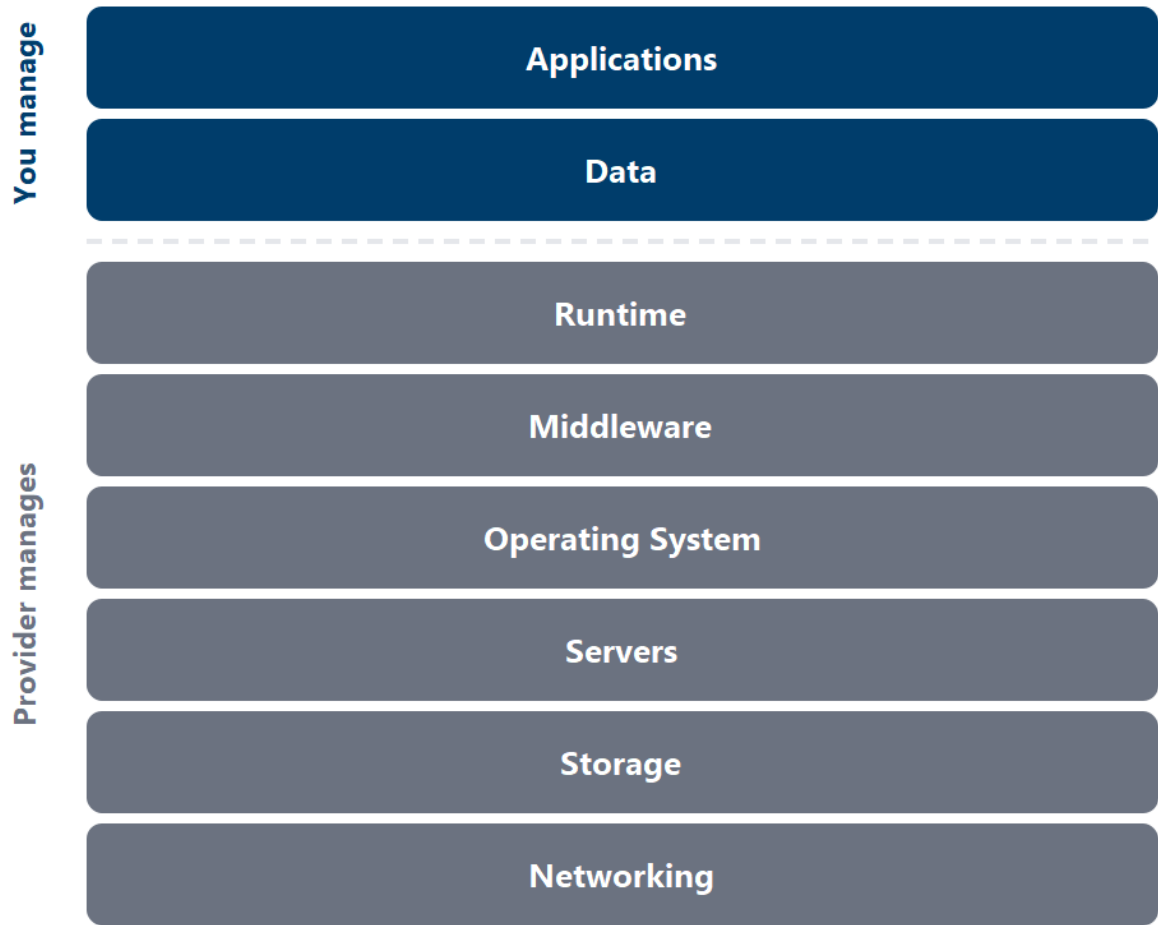
Rapidly replicate dev/test environments. Start up or shut down with complete control.

Key takeaway

You rent hardware — what you do with it is up to you.

Platform as a service (PaaS)

Balanced — you focus on your application and data



Common Scenarios

Development framework

Build cloud-based applications using built-in components with scalability and high availability.

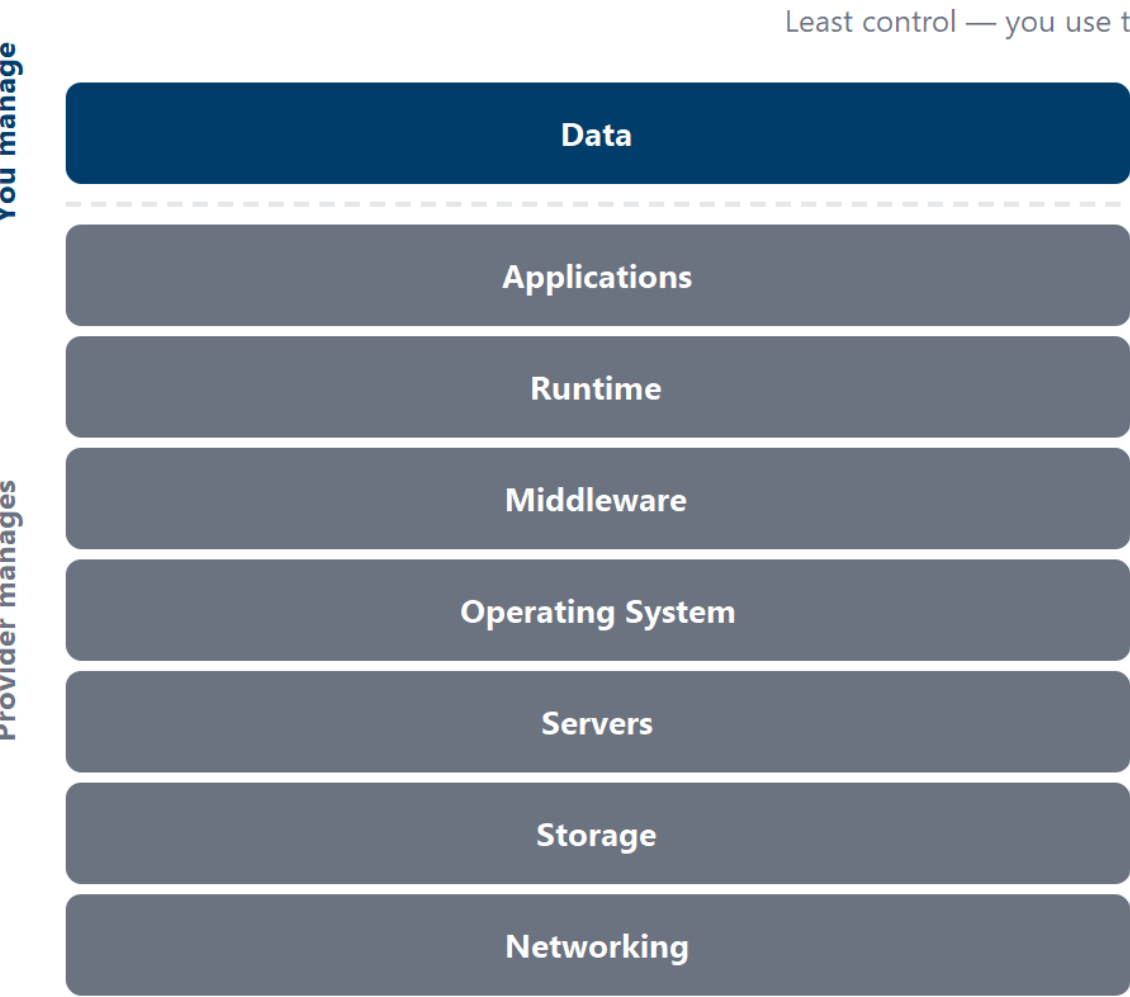
Analytics and business intelligence

Analyze and mine data, find insights, and predict outcomes using platform tools.

Key takeaway

You build on the platform — no infrastructure headaches.

Software as a service (SaaS)



Least control — you use the fully managed application

Common Scenarios

Email and messaging

Microsoft 365, Teams, Outlook

Productivity applications

Word, Excel, PowerPoint online

Finance and expense tracking

Cloud-based financial management tools

Key takeaway

You use the app — the provider handles everything else.

Knowledge check



1 Which service type gives the customer the most control over OS configuration and patching?

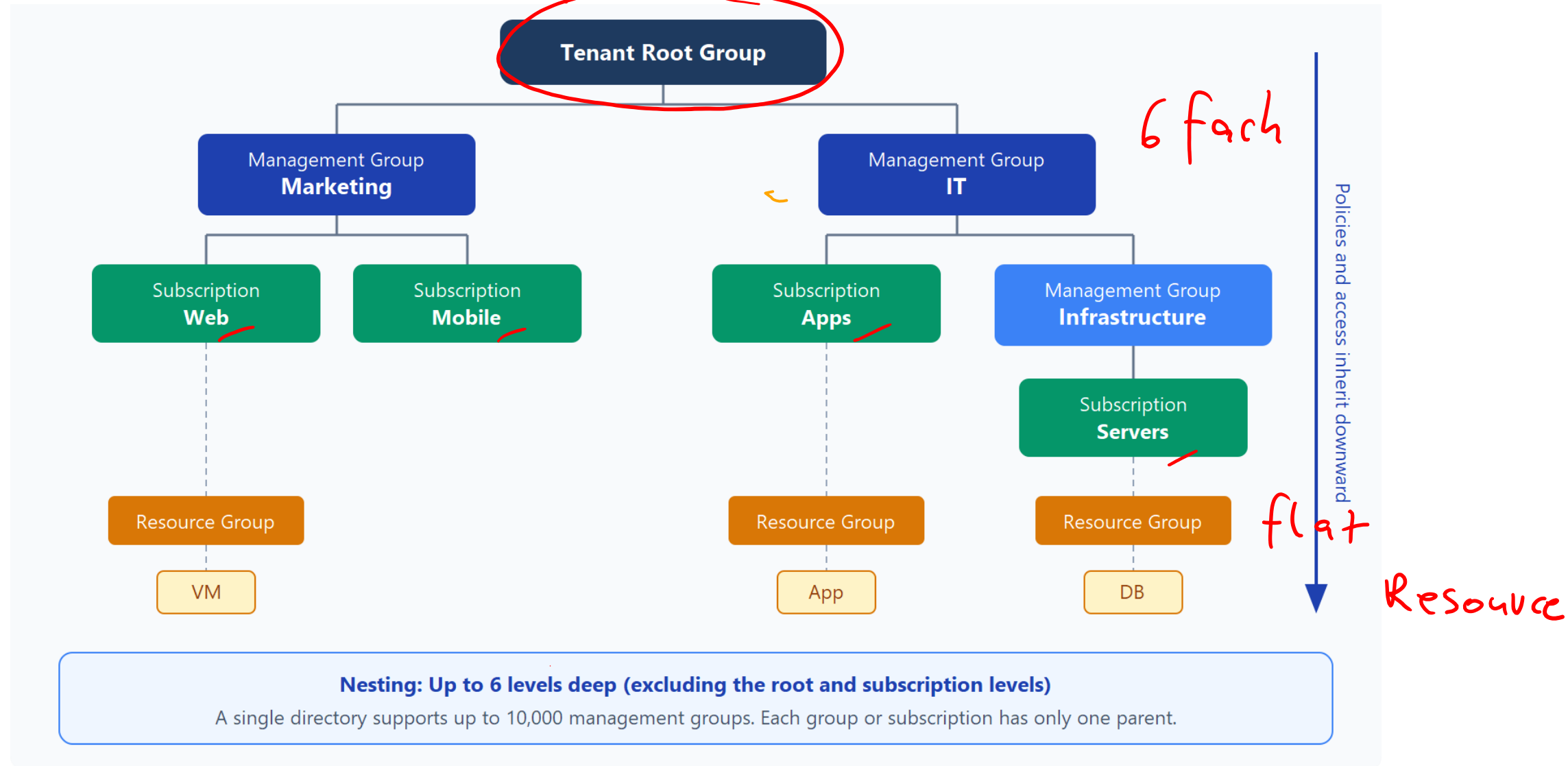
- ☐ Infrastructure as a Service (IaaS)
- ☐ Platform as a Service (PaaS)
- ☐ Software as a Service (SaaS)

Describe the core architectural components of Azure



Azure management infrastructure

Entr ID

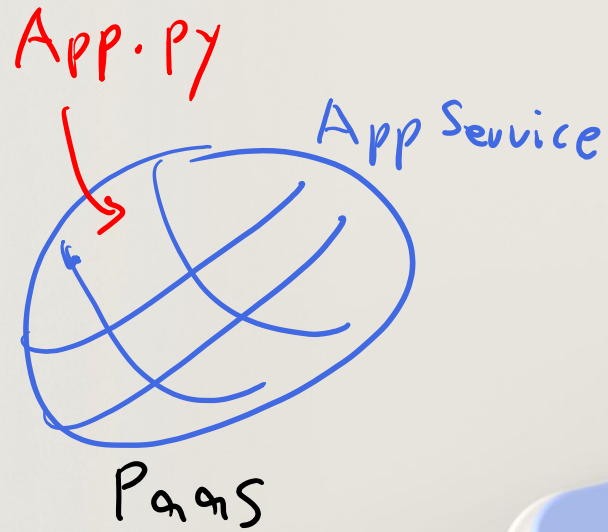




IaaS



CaaS



Describe Azure compute services



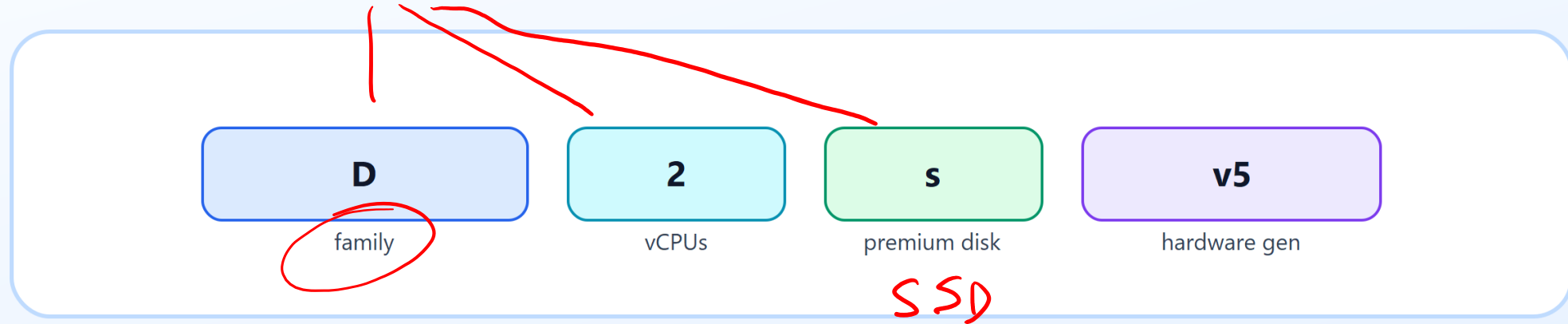
Decoding Azure Virtual Machine names

Size

T - Shirt

S M L XL XXL

How to read Standard_D2s_v5



Azure Virtual Machines families

VM size families

Choose the family that best matches workload behavior.

B

burstable

D

general

E

memory

F

compute

M

large memory

L

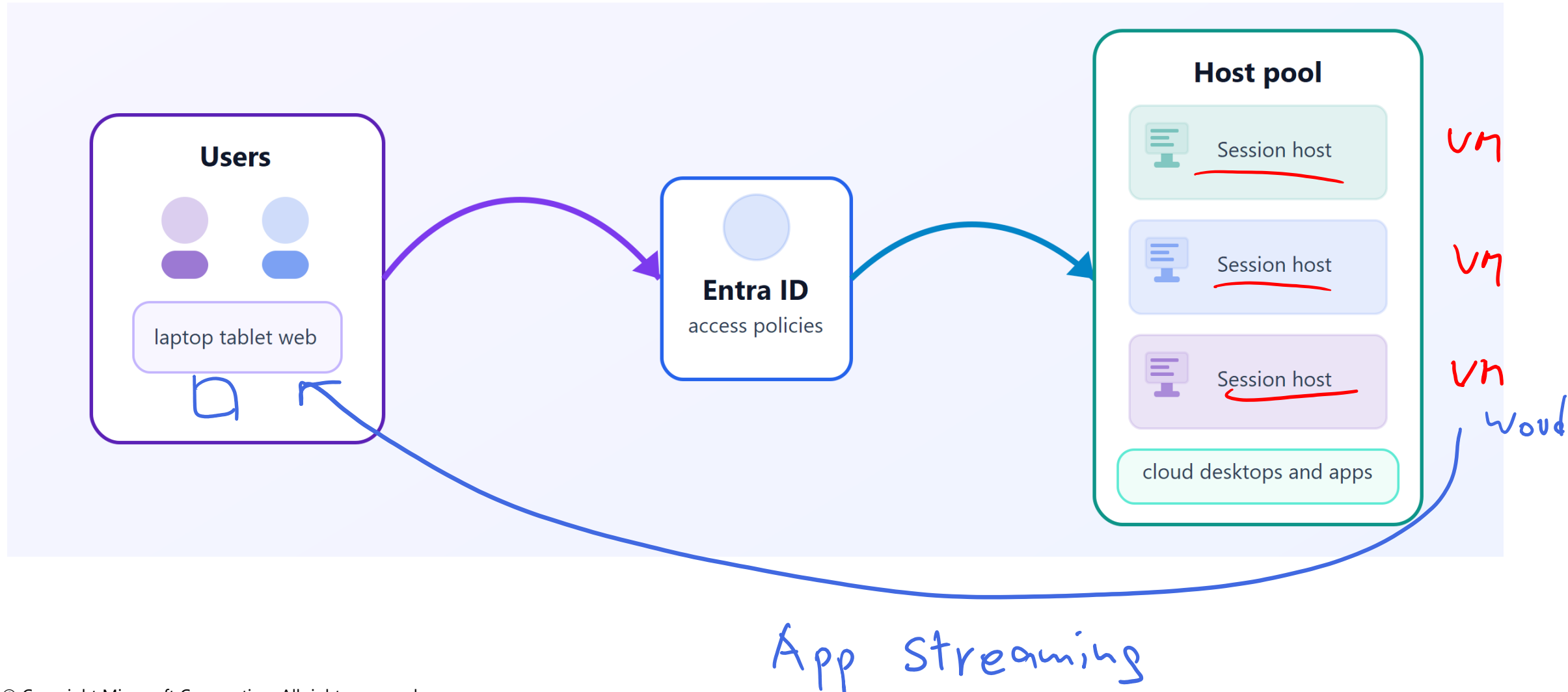
storage

N

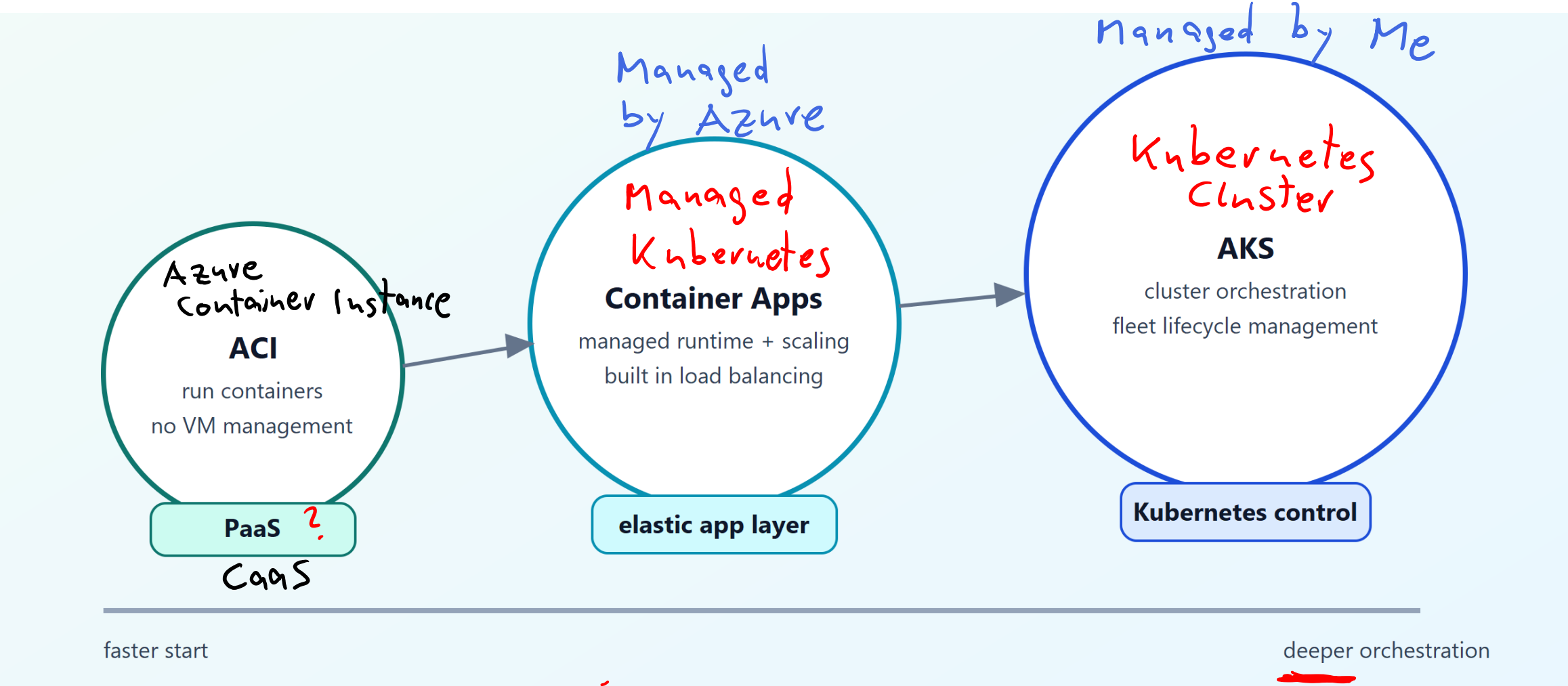
GPU

Azure Virtual Desktop

AVD = Citrix Light

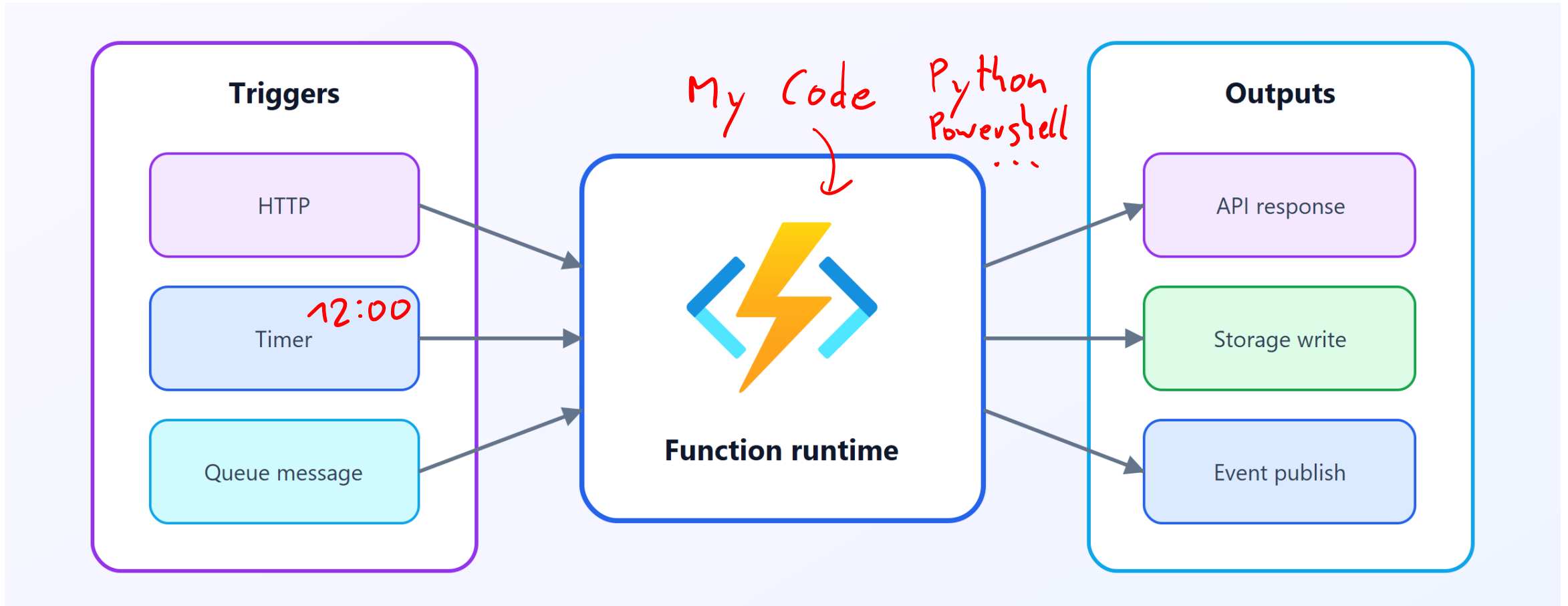


Azure container services

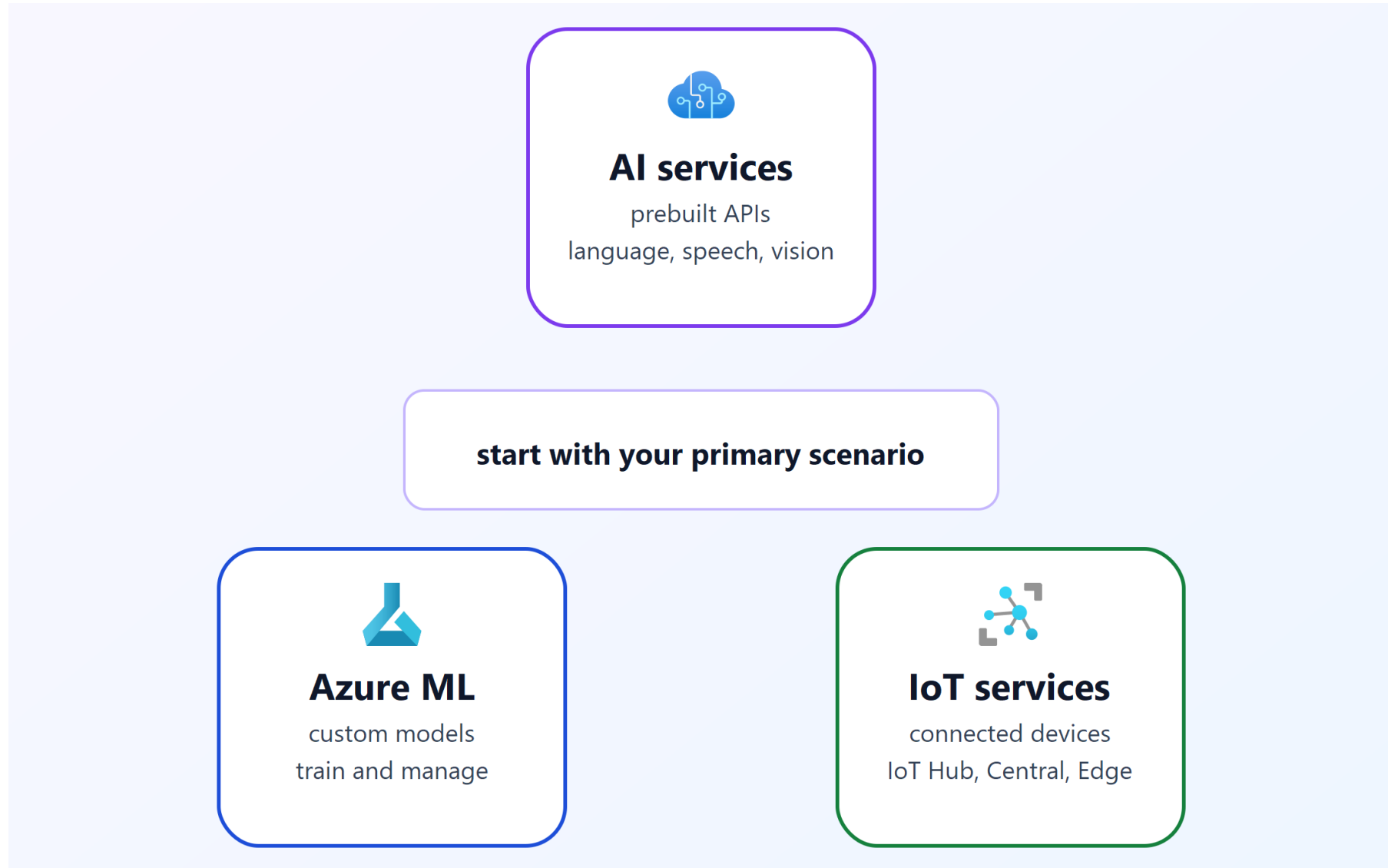


Serverless Compute

Azure Functions



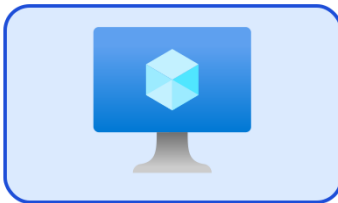
AI, machine learning, and IoT and Edge services



Application hosting options

Virtual Machines

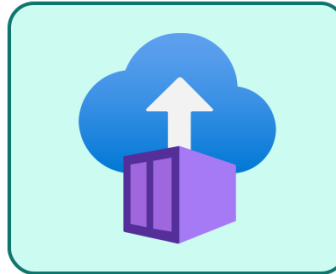
OS and config control
custom hosting setup
you patch and maintain



more control

Containers

component isolation
portable deployments
independent scaling



App Service

managed platform
autoscale and HA
integrated deployment



less operational effort

Describe Azure storage services



Azure storage accounts

Windows 18
C:\temp
Share
SMB

Anatomy of an Azure Storage Account



Account Name: Unique Namespace ←

3-24 chars, lowercase + numbers, globally unique in Azure

contains

STORAGE SERVICES



Blobs

Objects, Data Lake
.blob / .dfs



Files

SMB / NFS shares
.file



Queues

Async messaging
.queue



Tables

NoSQL structured
.table

protected by

REDUNDANCY OPTIONS

LRS

11 nines

ZRS

12 nines

GRS

16 nines

GZRS

16 nines

RA-GRS

16 nines

RA-GZRS

16 nines

A storage account provides a unique namespace, contains one or more services, and is protected by a redundancy option.

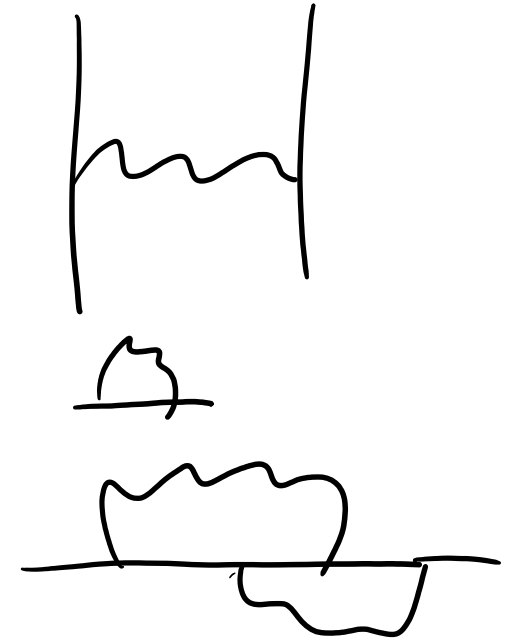
Binary
Large
Object

Container
für Blobs

Azure storage redundancy

- LRS replicates data 3x in a single datacenter (11 nines durability); lowest cost
- ZRS replicates across 3 availability zones in the primary region (12 nines durability)
- GRS uses LRS in both primary and secondary regions (16 nines durability)
- RA-GRS and RA-GZRS enable read access to secondary region data before failover

Richard Feynman
Nobel 1963 QED



Azure storage services

Azure Storage Services

Azure Blobs

Object store for text and binary data
Includes Data Lake Storage Gen2

Azure Files

Managed file shares via SMB/NFS
Cloud and on-premises access

Azure Queues

Messaging store for async processing
Messages up to 64 KB each

Azure Disks

Block-level storage volumes for VMs
Fully managed by Azure

Azure Tables

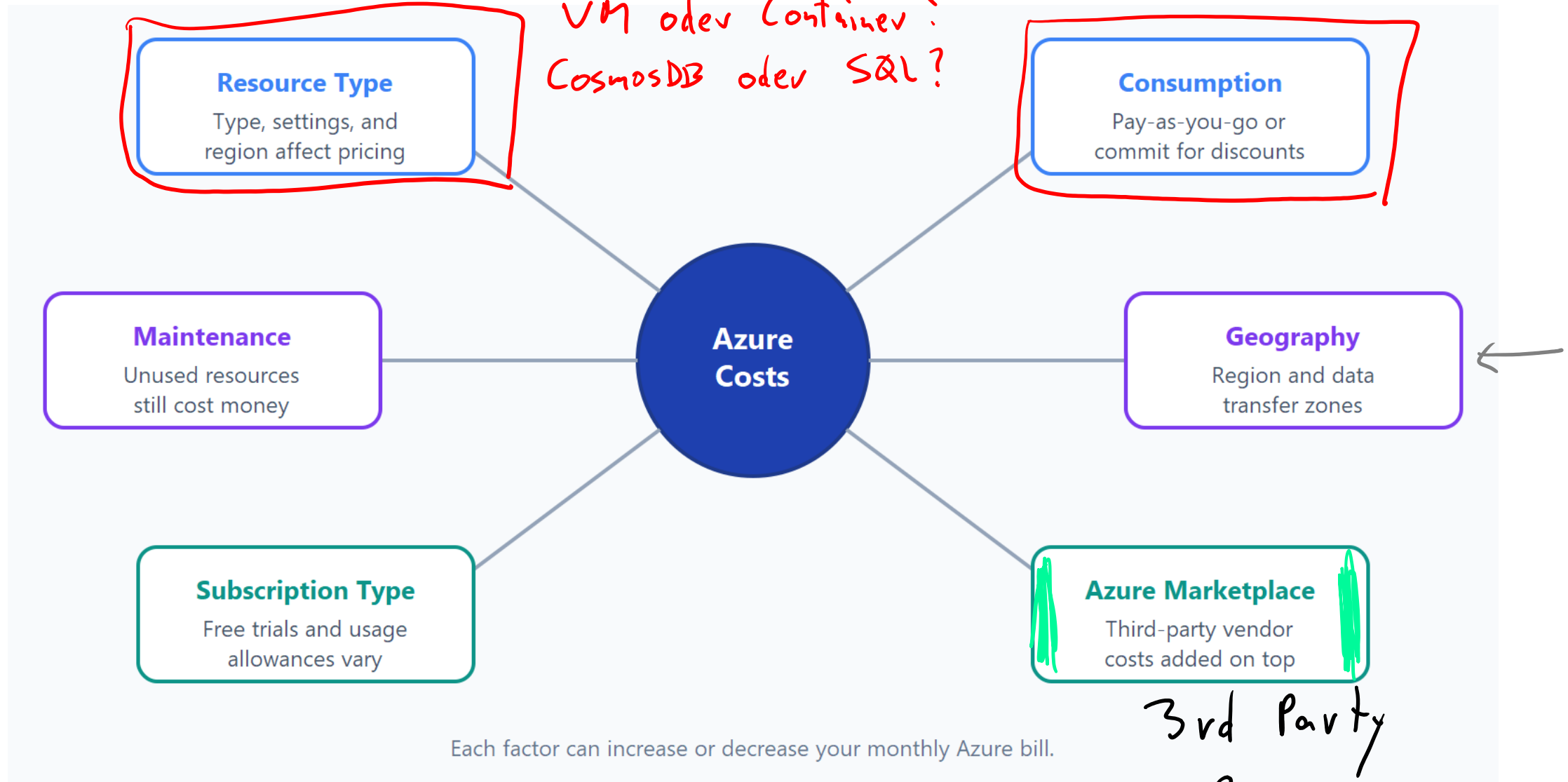
NoSQL store for structured data
Non-relational data storage

Describe cost management in Azure

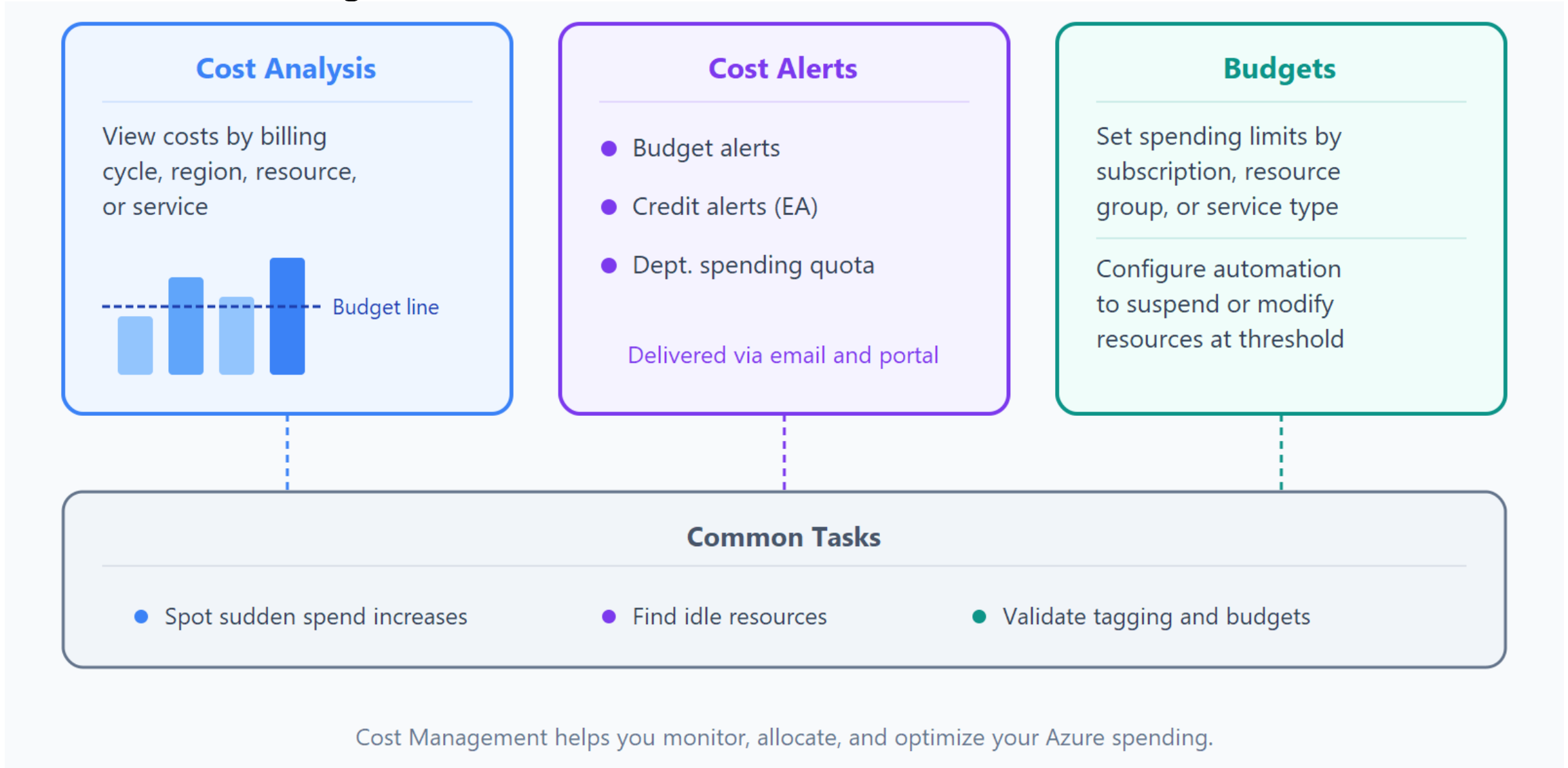


Price Calculator

Factors that affect costs in Azure



Microsoft Cost Management



Cost optimization options

- Reservations: commit to specific resources for 1 or 3 years at discounted pricing
- Savings plan for compute: commit to hourly spend amount across eligible compute services
- Spot Virtual Machines: use unused Azure capacity at reduced prices for interruptible workloads
- Decision guide: Reservations for stable workloads, savings plans for flexible compute, Spot for fault-tolerant jobs
- Review choices regularly as workload patterns change over time

Knowledge check



1 Which pricing option is usually best for steady, predictable workloads over a 1- or 3-year term?

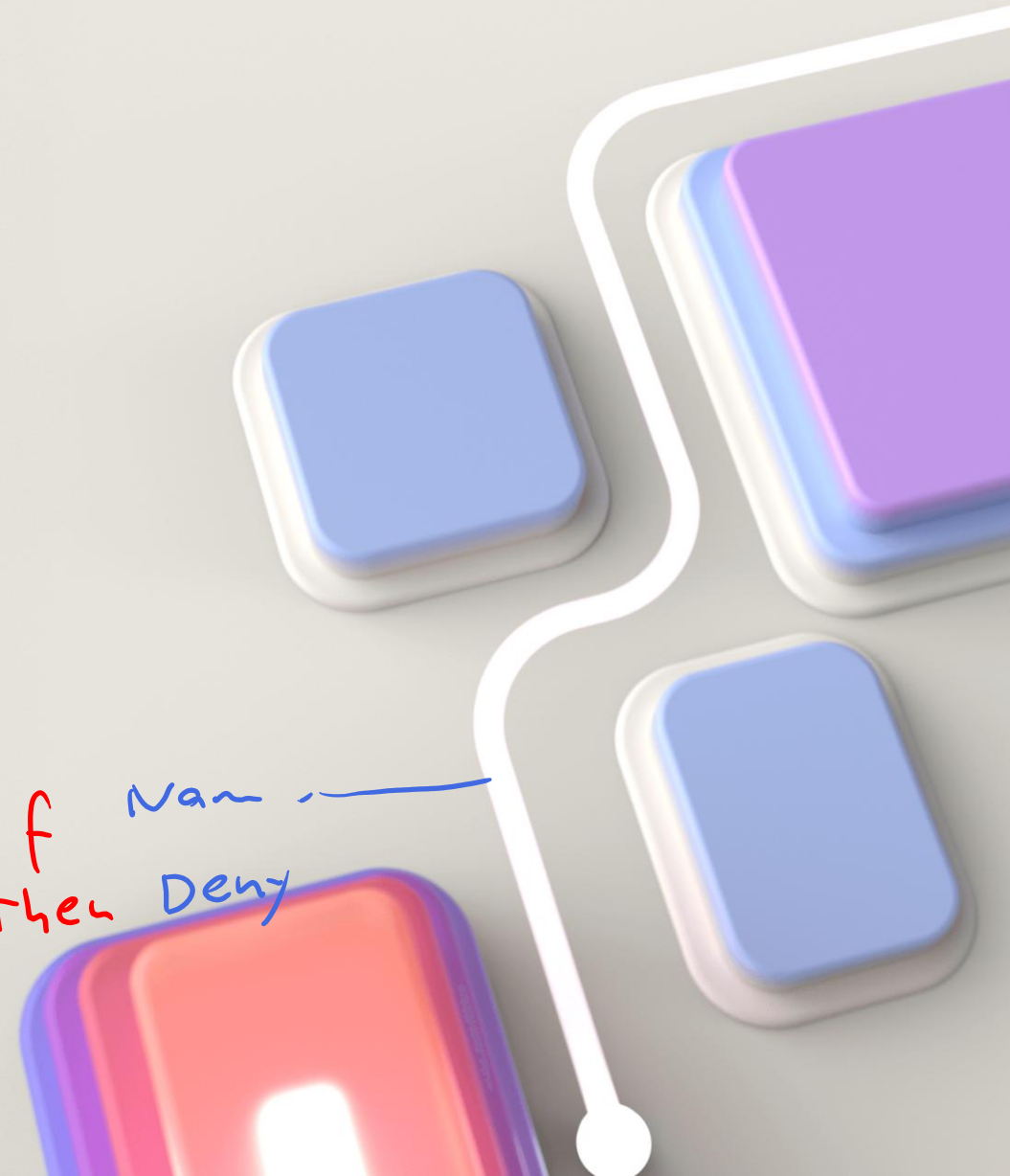
- ☐ Spot VMs
- ☐ Reservations
- ☐ Pay-as-you-go only

Describe features and tools for governance and compliance

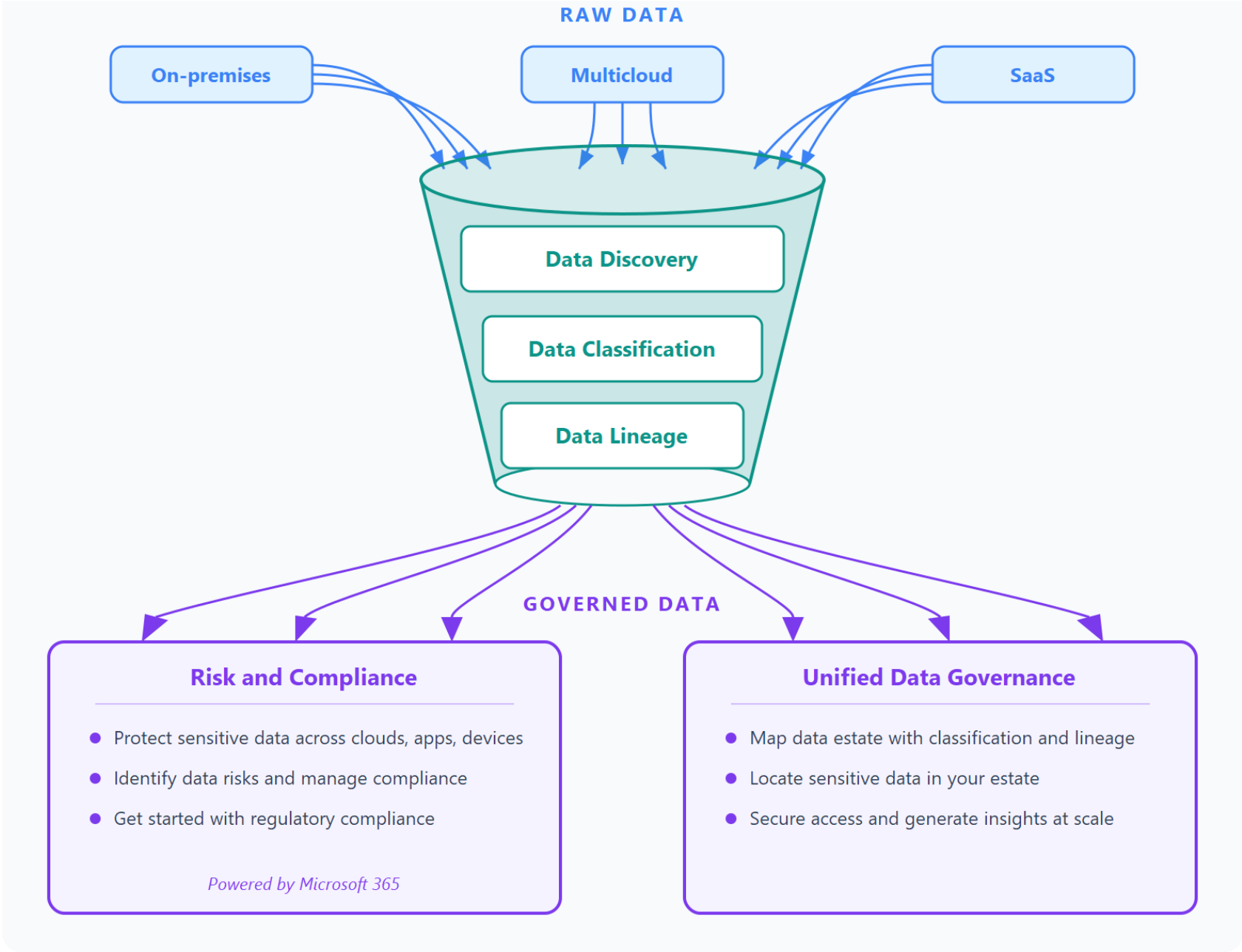
Policies
json Definition

if then Deny

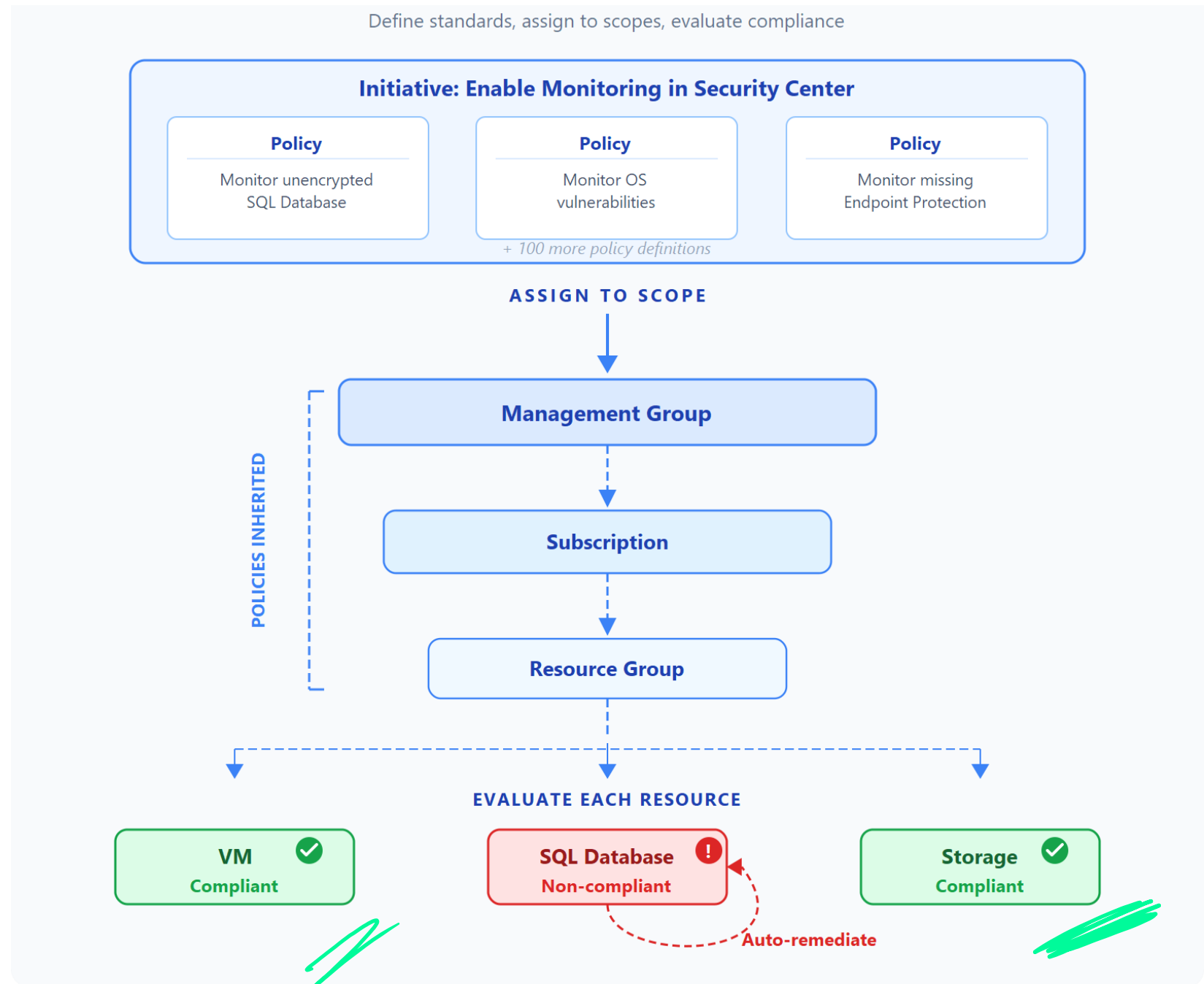
name



Microsoft Purview



Azure Policy



Resource locks

LOCK TYPES



Delete Lock

Read	✓ Allowed	Delete	✗ Blocked
Modify	✓ Allowed		



ReadOnly Lock

Read	✓ Allowed	Delete	✗ Blocked
Modify	✗ Blocked		

Modifying a Locked Resource

- 1 Remove the lock →
- 2 Perform the action

*Locks apply regardless of RBAC permissions.
Even an Owner must remove the lock first.*

INHERITANCE



Resource Group

inherits



Virtual Machine



Storage Account



SQL Database

All resources inherit the lock

Knowledge check



1 Which service is used to enforce standards and evaluate resource compliance at scale?

- ☒ Azure Policy
- ☐ Azure Advisor
- ☐ Azure Monitor

*
Fundamental

**
Associate

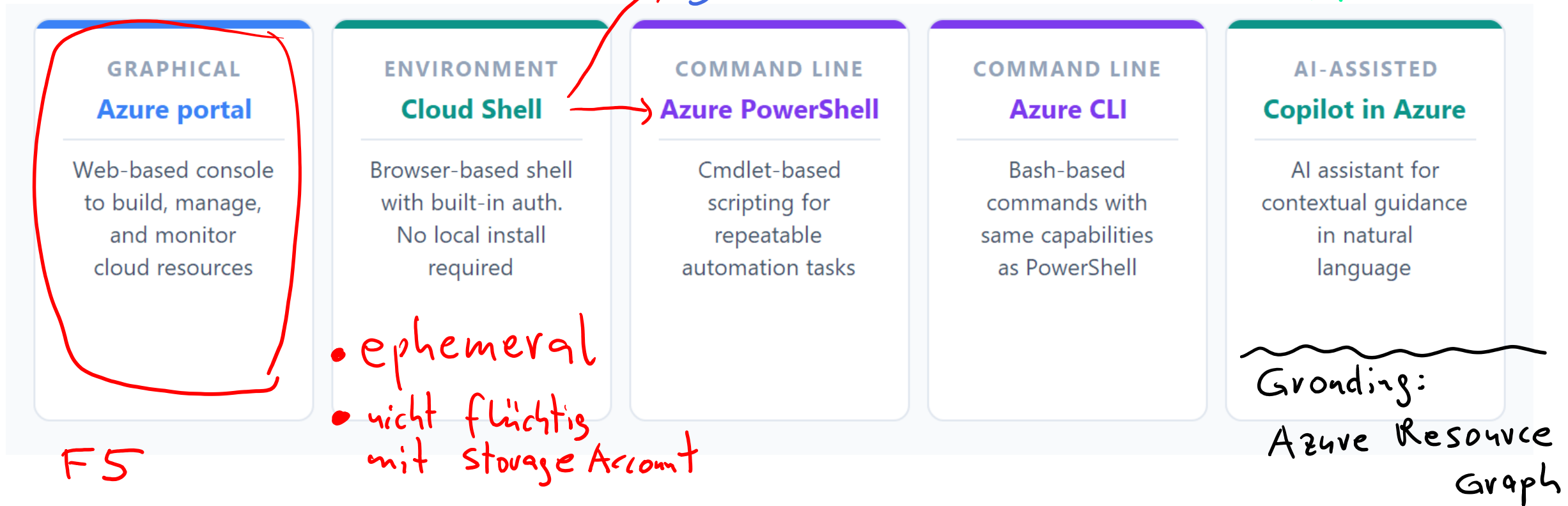
Expert



Describe features and tools
for managing and deploying
Azure resources



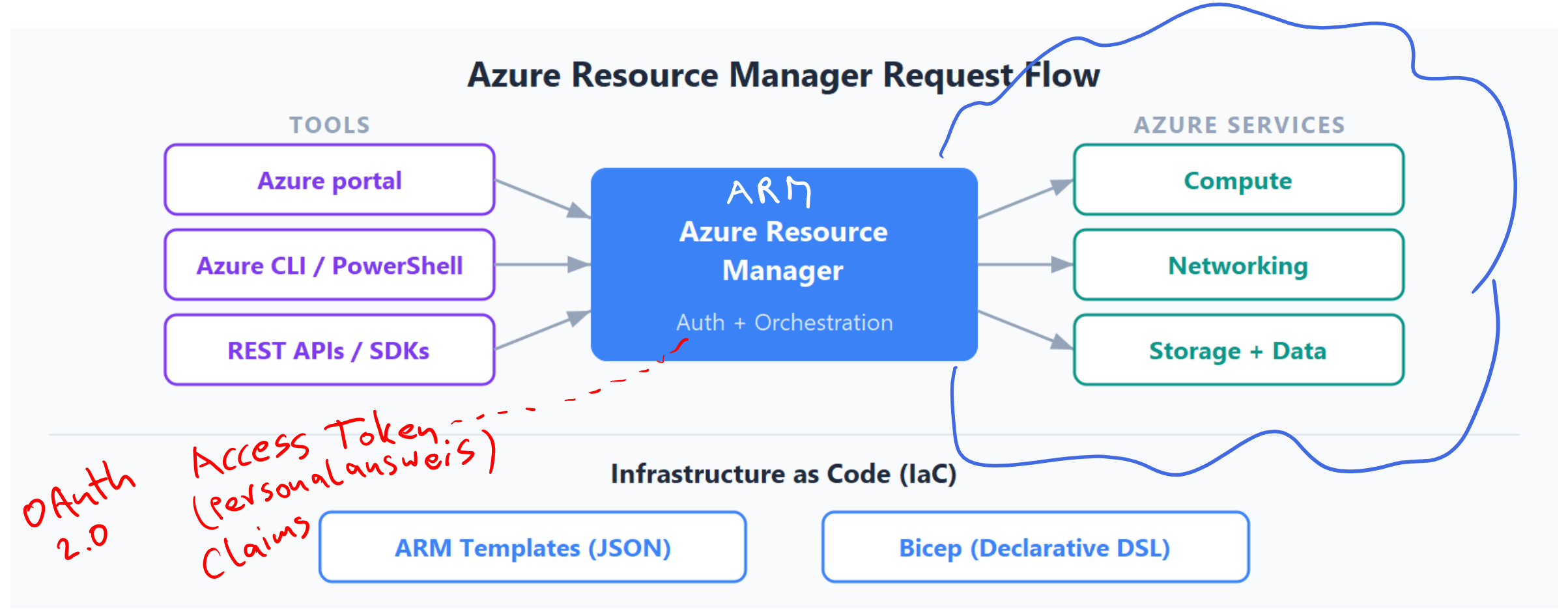
Tools for interacting with Azure



<https://portal.azure.com>

Azure Resource Manager, ARM templates, and Bicep

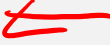
Azure Resource Manager Request Flow



Knowledge check



1 Which approach is best aligned to repeatable, version-controlled Azure deployments?

- ☐ Manual portal changes only
- ☒ Infrastructure as Code (for example, ARM/Bicep) 
- ☐ Ad-hoc scripts with no source control

End of presentation



